COSC 236 GRASP Report

Information Expert

The Information Expert Principle was applied to the Book, Member, and Library classes.

The Book class knows its own availability, updating its own status using borrow() and returnBook().

The Member class knows what books it has borrowed, keeping track of a list of borrowed books using borrowBook() and returnBook().

The Library class knows what members have registered and what books are catalogued, retrieving information from both lists using findBookByTitle() and findMember().

Creator

The Library class applies the Creator principle by creating Book objects in addBook() and Member objects in registerMember(). Library stores and manages these objects in its catalogue and registered-member list. It also supports removal through removeBook() and removeMember().

Controller

The LibrarianController class handles borrowing and returning requests through borrowBook() and returnBook(). It delegates these requests to Library, which locates the appropriate Member and Book. Member manages its borrowed-books list, while Book updates its own availability.

Low Coupling & High Cohesion

Each class interacts with the others through public methods instead of directly modifying their attributes, promoting Low Coupling. Each class also works on one responsibility (Book tracks its availability, Member tracks borrowed books, Library tracks what books/members are part of the system, LibrarianController handles requests), promoting High Cohesion

Application Demonstration

LibraryApp contains main() and demonstrates the system by calling Library and LibrarianController rather than maintaining separate book and member collections.
The demonstration uses Alice and Bob, and the books Dune, 1984, and Moby Dick. It covers adding and removing books and members, looking up a book and a member, displaying available books, borrowing, and returning.
It also demonstrates rejected requests: borrowing an unavailable book, returning a book the member does not hold, removing a borrowed book, and removing a member with outstanding loans. These checks help preserve consistency between the catalogue, book availability, and members' borrowed-books lists.